

EXPT: 7 — SLIDING WINDOW PROTOCOL

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AIM

To implement flow control at the data link layer using Sliding Window Protocol and simulate the flow of frames from sender to receiver.

PROCEDURE

1. Start the program.
2. Enter the window size and text message.
3. Divide the message into frames, one character per frame.
4. Display the frames in the sender buffer.
5. Send frames in the sequence according to the window size.
6. Receive ACK from the receiver.
7. If the correct ACK is received, slide the window and send the next frames.
8. If NACK is received, retransmit the frames.
9. Continue until all frames are transmitted.
10. Stop the program.

PROGRAM

Sender.py

```
import time

import os

window_size = int(input("Enter window size: "))

message = input("Enter text message: ")

frames = []

for i, char in enumerate(message):

    frames.append((i, char))

next_frame = 0

while next_frame < len(frames):

    end = min(next_frame + window_size, len(frames))

    with open("sender_buffer.txt", "w") as file:
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```

    for i in range(next_frame, end):

        file.write(f"Frame No: {frames[i][0]} Data: {frames[i][1]}\n")

print("\nFrames sent:")

for i in range(next_frame, end):

    print(f"Frame No: {frames[i][0]} Data: {frames[i][1]}")

print("\nWaiting for acknowledgement...")

time.sleep(3)

while True:

    if os.path.exists("receiver_buffer.txt"):

        with open("receiver_buffer.txt", "r") as file:

            ack = file.read().strip()

            if ack:

                break

        time.sleep(1)

print("Receiver response:", ack)

if ack.startswith("ACK"):

    ack_no = int(ack.split(":")[1].strip())

    if ack_no == end:

        print(f"ACK {ack_no} received")

        print("Sliding window moved forward")

        next_frame = end

    else:

        print("Unexpected ACK")

        print("Resending current window")

```

Receiver.py

```
import time

import os

expected_frame = 1

print("Receiver started")

while True:

    if not os.path.exists("sender_buffer.txt"):

        time.sleep(1)

        continue

    with open("sender_buffer.txt", "r") as file:

        data = file.readlines()

    if not data:

        time.sleep(1)

        continue

    print("\nReceiver Frames:")

    for line in data:

        print(line.strip())

    for line in data:

        parts = line.strip().split()

        frame_no = int(parts[2])

        frame_data = parts[4]

        if frame_no == expected_frame:

            print(f"Frame {frame_no} received: {frame_data}")

            with open("receiver_buffer.txt", "w") as file:

                file.write(f"ACK:{frame_no}")

            expected_frame += 1
```

```
else:

    print(f"Error: Expected frame {expected_frame}, but received frame {frame_no}")

    print(f"NACK {expected_frame} sent")

    with open("receiver_buffer.txt", "w") as file:

        file.write(f"NACK:{expected_frame}")

    time.sleep(1)
```

ERROR TEST / OUTPUT

The handwritten test output shows:

Error: Expected frame 2, but received frame 5

NACK 2 sent

Sender O/P:

Receiver response: NACK:2

NACK 2 received

Resending frames from the error frame.

INPUT

Enter window size: 3

Enter text message: Hello

OUTPUT — SENDER

Sender:

Frames sent:

Frame No: 1 Data: H

Frame No: 2 Data: E

Frame No: 3 Data: L

Waiting for acknowledgement...

Receiver response: ACK:3

ACK 3 received

Sliding window moved forward

OUTPUT — RECEIVER

Receiver:

Receiver started

Received frames:

Frame No: 1 Data: H

ACK 3 sent

The second page continues with the error/retransmission test.

Error Test Output

Error test O/P:

Receiver O/P:

Error: Expected frame 2, but received frame 5

NACK 2 sent

Sender O/P:

Receiver response: NACK:2

NACK 2 received

Resending frames from the error frame.